

Math 105: Calculus with Algebra
Instructor: David Zureick-Brown (“DZB”)

All assignments

Gradescope code: **B5J8EN**

Gradescope instructions	2
1 Assignment 1, Due Friday, September 4	4
2 Assignment 1 Solutions	6
3 Assignment 2, Due Monday, September 7	9
4 Assignment 2 Solutions	10
5 Assignment 3, Due Friday, September 11	13
6 Assignment 3 Solutions	16
7 Assignment 4, Due Monday, September 14	21
8 Assignment 4 Solutions	23
9 Assignment 5, Due Friday, September 19	27
10 Assignment 5 Solutions	29
11 Assignment 6, Due Monday, September 21	34
12 Assignment 6 Solutions	36
13 Assignment 7, Due Friday, September 26	40

Gradescope Instructions for submitting work in Math 105

You will be using the online Gradescope program to submit your homework and exams. These instructions tell you how to sign up initially, and how to submit your written work.

Signing up for Gradescope the first time.

If you haven't used Gradescope for an **Amherst College** course before:

- Go to <http://www.gradescope.com>, click on "Sign up for free" (which may auto-scroll you to the bottom of the page), and select Sign up as [a] "Student".
- In the signup box:
 - Use the course entry code **B5J8EN**
 - Use your full name
 - Use your **Amherst College email** address. Or, if you are a Five-College student, use your email address from your own school.
 - Leave the "Student ID" entry blank.
- You will probably get an email asking to set a password for your account, so check your **amherst.edu** email inbox. (Or your email inbox through your own school, for Five-College students.)

Adding Math 105 to Gradescope.

If you **have** used Gradescope for an Amherst course before, and so you already have an account through your **amherst.edu** email, you still need to add Math 105, so:

- Go to <http://www.gradescope.com> and log in.
- Go to your Account Dashboard (click the Gradescope logo at upper left), and click "Add Course" at bottom right.
- Use the course code **B5J8EN**

(submission instructions on next page)

Submitting written work

First write it out on paper as you would normally. Then **scan it** to create a PDF. One method for scanning is the smartphone app **DropBox**. It makes nice clear scans, and it saves them directly into a folder so that you can have all your assignments in one place. **CamScanner** is another free scanning App, and there are others, too. **Gradescope** now has its own scanning app. You can also use a printer/scanner if you prefer.

Please be kind to our dear graders and make sure your submission is legible !

In particular, please leave some spacing between separate problems.

If you have a tablet computer, you may write your work there (instead of on paper) and save it as a PDF.

Some of you may know the math formatting package LaTeX and may want to use it in Math 105. That's fine, too; if so, you may write up your work in LaTeX and save the resulting PDF.

In short, any method is fine as long as it creates a legible PDF file and NOT a photo.

For example, if you use the DropBox app, then in your created *Math 105 Homework* Dropbox folder, you can select create (+) at the bottom of the screen and click the *Scan Document* option. Snap a shot of the first page of your homework, and then click [+] to snap shots of any subsequent pages. Do **not** use the *Take Photo* option.

After you have scanned/saved your work as a PDF, submit it on Gradescope as follows:

- Go to <http://www.gradescope.com> and log in.
- Select the course “Math 105, Fall 2026” and the appropriate assignment.
- Select “submit pdf” to submit your work in PDF format. Browse to find your PDF and upload.
- Now it is time to **tag** your problems. **This is an important step**, where you are telling Gradescope which problems are on which page(s).

For each problem, select the pages of your submission where your written solution appears.

I think the easiest thing to do is to click on the page of **your** homework upload where you wrote the given problem, and then click on the assigned problem listed. Repeat for each problem.

You must tag the problems or else you will not get credit for your work.

Gradescope will give you a warning when you go to submit your assignment if you have not selected the pages correctly. If you tag a problem incorrectly, you can fix it by clicking “More” and “Reselect Pages”.

- Click Save or Submit.

After your assignment is graded, you will be able to see your score on the written problems, along with comments, on Gradescope. You should receive an email notifying you when each homework set is graded.

MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 1

Due **Friday, September 4** at 11:55pm, on Gradescope

GOALS:

- Review of basic algebra.
- Review of absolute values and inequalities.
- Write complete solutions.

Justify your solution, writing down the steps that led you to the final answer. For a review on inequalities and absolute value, see Appendix A (Stewart Calculus book).

(1) Simplify:

$$4 - (5 - 3) + 3(4 - 7) - 4(1 + 2)$$

(2) In the following expressions, simplify and reduce to lowest terms. (Some answers may have more than one form)

(a)

$$\frac{\frac{4}{75}}{\frac{8}{25}}$$

(b)

$$\frac{7x}{3y} \cdot \frac{3y+2}{x}$$

(c)

$$\frac{\frac{xy}{(x+y)}}{\frac{x^2y}{(x+y)^3}}$$

MORE PROBLEMS ON THE NEXT PAGE →

- (3) Express as a single fraction and simplify:

(a)

$$\frac{1}{2} + \frac{4}{3} - \frac{2}{5} - \frac{3}{15}$$

(b)

$$\frac{\frac{1}{3} + \frac{2}{5}}{\frac{3}{2}}$$

(c)

$$\frac{1}{x} - \frac{x+1}{xy} + \frac{x-2}{xz}$$

(d)

$$\frac{\frac{1}{st} - \frac{1}{w}}{\frac{1}{tw} - \frac{2}{s}}$$

- (4) Rewrite the expression without using the absolute-value symbol:

$$|x - 2| \text{ if } x < 2$$

- (5) Solve the inequalities with the **geometric interpretation** of absolute value.

(a)

$$|x| < 3$$

(b)

$$|x - 4| < 1$$

(c)

$$|x + 5| \geq 2$$

MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 1 SOLUTIONS

- (1) Simplify:

$$4 - (5 - 3) + 3(4 - 7) - 4(1 + 2)$$

Solution. We simplify the expressions inside parentheses first:

$$4 - (5 - 3) + 3(4 - 7) - 4(1 + 2) = 4 - 2 + 3(-3) - 4(3).$$

Therefore

$$4 - 2 - 9 - 12 = -19.$$

Hence the simplified value is $\boxed{-19}$.

- (2) In the following expressions, simplify and reduce to lowest terms. (Some answers may have more than one form)

- (a)

$$\frac{\frac{4}{75}}{\frac{8}{25}}$$

Solution. Dividing by a fraction is the same as multiplying by its reciprocal:

$$\frac{\frac{4}{75}}{\frac{8}{25}} = \frac{4}{75} \cdot \frac{25}{8} = \frac{1}{3} \cdot \frac{1}{2} = \boxed{\frac{1}{6}}.$$

- (b)

$$\frac{7x}{3y} \cdot \frac{3y+2}{x}$$

Solution. Cancel the common factor x :

$$\frac{7x}{3y} \cdot \frac{3y+2}{x} = \boxed{\frac{7(3y+2)}{3y}}.$$

- (c)

$$\frac{\frac{xy}{(x+y)}}{\frac{x^2y}{(x+y)^3}}$$

Solution. Multiply by the reciprocal of the denominator:

$$\frac{\frac{xy}{(x+y)}}{\frac{x^2y}{(x+y)^3}} = \frac{xy}{x+y} \cdot \frac{(x+y)^3}{x^2y}.$$

Cancelling common factors gives

$$\boxed{\frac{(x+y)^2}{x}}.$$

- (3) Express as a single fraction and simplify:

(a)

$$\frac{1}{2} + \frac{4}{3} - \frac{2}{5} - \frac{3}{15}$$

Solution. A common denominator is 30:

$$\frac{1}{2} + \frac{4}{3} - \frac{2}{5} - \frac{3}{15} = \frac{15}{30} + \frac{40}{30} - \frac{12}{30} - \frac{6}{30} = \boxed{\frac{37}{30}}.$$

(b)

$$\frac{\frac{1}{3} + \frac{2}{5}}{\frac{3}{2}}$$

Solution. First simplify the numerator:

$$\frac{1}{3} + \frac{2}{5} = \frac{5}{15} + \frac{6}{15} = \frac{11}{15}.$$

Then divide by $3/2$:

$$\frac{\frac{11}{15}}{\frac{3}{2}} = \frac{11}{15} \cdot \frac{2}{3} = \boxed{\frac{22}{45}}.$$

(c)

$$\frac{1}{x} - \frac{x+1}{xy} + \frac{x-2}{xz}$$

Solution. The common denominator is xyz . Thus

$$\frac{1}{x} - \frac{x+1}{xy} + \frac{x-2}{xz} = \frac{yz}{xyz} - \frac{z(x+1)}{xyz} + \frac{y(x-2)}{xyz}.$$

Combining the numerators,

$$\boxed{\frac{xy + yz - xz - 2y - z}{xyz}}.$$

(d)

$$\frac{\frac{1}{st} - \frac{1}{w}}{\frac{1}{tw} - \frac{2}{s}}$$

Solution. Put the numerator and denominator over the common denominator stw :

$$\frac{1}{st} - \frac{1}{w} = \frac{w-st}{stw}, \quad \frac{1}{tw} - \frac{2}{s} = \frac{s-2tw}{stw}.$$

Therefore

$$\frac{\frac{w-st}{stw}}{\frac{s-2tw}{stw}} = \boxed{\frac{w-st}{s-2tw}}.$$

(4) Rewrite the expression without using the absolute-value symbol:

$$|x-2| \text{ if } x < 2$$

Solution. If $x < 2$, then $x-2 < 0$. The absolute value of a negative number is its opposite, so

$$|x-2| = -(x-2) = \boxed{2-x}.$$

(5) Solve the inequalities with the **geometric interpretation** of absolute value.

(a)

$$|x| < 3$$

Solution. The quantity $|x|$ is the distance from x to 0. We want that distance to be less than 3, so x must lie between -3 and 3 :

$$\boxed{-3 < x < 3}.$$

(b)

$$|x - 4| < 1$$

Solution. The quantity $|x - 4|$ is the distance from x to 4. We want that distance to be less than 1, so

$$\boxed{3 < x < 5}.$$

(c)

$$|x + 5| \geq 2$$

Solution. Since $|x + 5| = |x - (-5)|$, this is the distance from x to -5 . We want that distance to be at least 2. Thus x must be at least 2 units to the left or right of -5 :

$$\boxed{x \leq -7 \quad \text{or} \quad x \geq -3}.$$

MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 2

Due **Monday, September 7** at 11:55pm, on Gradescope

GOALS:

- Review equations of lines.
- Write complete solutions.

Justify your solution, writing down the steps that led you to the final answer. For a review on equations of lines, see Appendix A (Stewart Calculus book).

Find the slope of the line through P and Q :

(1) $P(1, 5), Q(4, 11)$

(2) $P(-1, -4), Q(6, 0)$

Find the equation of the line that satisfies the given conditions.

(3) Through $(2, -3)$ with slope 6

(4) Through $(2, 1)$ and $(1, 6)$

(5) Slope 3, y -intercept -2

(6) x -intercept 1, y -intercept -3

(7) Through $(4, 5)$, parallel to x -axis

(8) Through $(4, 5)$, parallel to y -axis

(9) Through $(1, -6)$, parallel to the line $x + 2y = 6$

(10) Through $(-1, -2)$, perpendicular to the line $2x + 5y + 8 = 0$

MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 2 SOLUTIONS

Find the slope of the line through P and Q :

- (1) $P(1, 5), Q(4, 11)$

Solution. The slope through two points (x_1, y_1) and (x_2, y_2) is

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

Thus

$$m = \frac{11 - 5}{4 - 1} = \frac{6}{3} = \boxed{2}.$$

- (2) $P(-1, -4), Q(6, 0)$

Solution. Using the slope formula,

$$m = \frac{0 - (-4)}{6 - (-1)} = \frac{4}{7}.$$

Therefore the slope is

$$\boxed{\frac{4}{7}}.$$

Find the equation of the line that satisfies the given conditions.

- (3) Through $(2, -3)$ with slope 6

Solution. Using point-slope form $y - y_1 = m(x - x_1)$,

$$y - (-3) = 6(x - 2).$$

Hence

$$y + 3 = 6x - 12,$$

so

$$\boxed{y = 6x - 15}.$$

- (4) Through $(2, 1)$ and $(1, 6)$

Solution. First find the slope:

$$m = \frac{6 - 1}{1 - 2} = \frac{5}{-1} = -5.$$

Using the point $(2, 1)$,

$$y - 1 = -5(x - 2).$$

Therefore

$$y - 1 = -5x + 10,$$

so

$$\boxed{y = -5x + 11}.$$

- (5) Slope 3, y -intercept -2

Solution. A line with slope m and y -intercept b has equation $y = mx + b$. Here $m = 3$ and $b = -2$, so

$$\boxed{y = 3x - 2}.$$

- (6) x -intercept 1, y -intercept -3

Solution. The x -intercept gives the point $(1, 0)$, and the y -intercept gives the point $(0, -3)$. The slope is

$$m = \frac{0 - (-3)}{1 - 0} = 3.$$

Since the y -intercept is -3 , the equation is

$$\boxed{y = 3x - 3}.$$

- (7) Through $(4, 5)$, parallel to x -axis

Solution. A line parallel to the x -axis is horizontal, so its y -coordinate is constant. Since the line passes through $(4, 5)$,

$$\boxed{y = 5}.$$

- (8) Through $(4, 5)$, parallel to y -axis

Solution. A line parallel to the y -axis is vertical, so its x -coordinate is constant. Since the line passes through $(4, 5)$,

$$\boxed{x = 4}.$$

- (9) Through $(1, -6)$, parallel to the line $x + 2y = 6$

Solution. Rewrite the given line in slope-intercept form:

$$x + 2y = 6 \implies 2y = -x + 6 \implies y = -\frac{1}{2}x + 3.$$

Its slope is $-\frac{1}{2}$. A parallel line has the same slope, so using point-slope form with $(1, -6)$,

$$y + 6 = -\frac{1}{2}(x - 1).$$

Therefore

$$\boxed{y = -\frac{1}{2}x - \frac{11}{2}}.$$

Equivalently, the equation can be written as $\boxed{x + 2y = -11}$.

- (10) Through $(-1, -2)$, perpendicular to the line $2x + 5y + 8 = 0$

Solution. First rewrite the given line in slope-intercept form:

$$2x + 5y + 8 = 0 \implies 5y = -2x - 8 \implies y = -\frac{2}{5}x - \frac{8}{5}.$$

Its slope is $-\frac{2}{5}$. A perpendicular line has slope equal to the negative reciprocal, so its slope is

$$\frac{5}{2}.$$

Using point-slope form with $(-1, -2)$,

$$y + 2 = \frac{5}{2}(x + 1).$$

Thus

$$y + 2 = \frac{5}{2}x + \frac{5}{2},$$

and therefore

$$\boxed{y = \frac{5}{2}x + \frac{1}{2}}.$$

MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 3

Due **Friday, September 11** at 11:55pm, on Gradescope

GOALS:

- Review of the essential ideas behind functions.
- Practice graphing of functions.
- Write complete solutions.

Justify your solution, writing down the steps that led you to the final answer.

(1) Let $f(x) = x^3 + 2x$. Determine:

(a) $f(2)$

(b) $f(2x)$

(c) $f(-x)$

(d) $f(2 + \Delta x)$

(2) Let $u(t) = \frac{1}{2}t^2 + t$. Determine:

(a) $u(-1)$

(b) $u(t + \Delta t)$

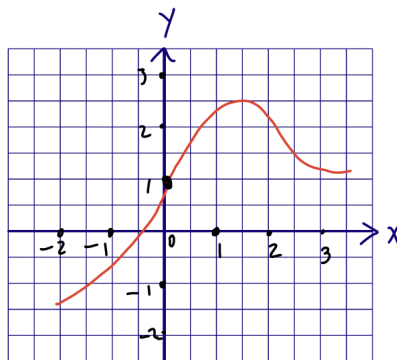
(c) $u(-t)$

(3) Sketch the graph of the function $g(t) = |1 - 3t|$

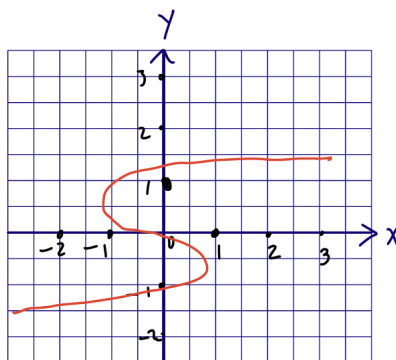
(4) Sketch the graph of the function

$$f(x) = \begin{cases} |x| & \text{if } |x| \leq 1 \\ 1 & \text{if } |x| > 1 \end{cases}$$

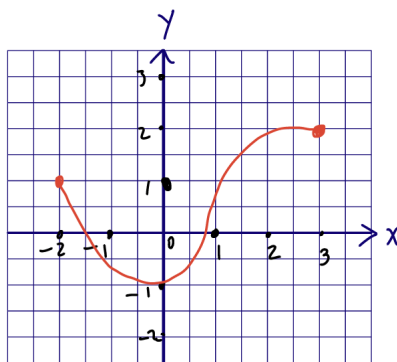
- (5) The graph of the function f is given.



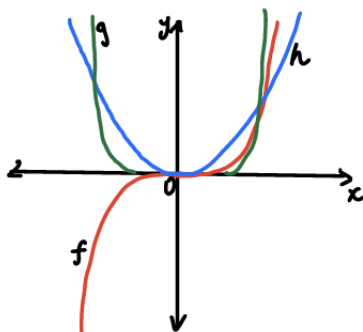
- (a) State the value of $f(1.5)$.
- (b) Estimate the value of $f(-1)$.
- (c) For what values of x is $f(x) = 1$?
- (d) Estimate the value of x such that $f(x) = 0$.
- (6) Determine whether the curve is a graph of a function of x . If it is, state the domain and the range of the function.



- (7) State the domain and the range of the function graphed below.



- (8) Find the domain of $f(x) = \frac{x+4}{x^2-9}$
- (9) Match the each equation with its graph. Explain your choices. Don't use a computer or graphic calculator. (a) $y = x^2$ (b) $y = x^5$ (c) $y = x^8$



MATH 105 Calculus with Algebra – Fall 2026

Homework 3 – Solutions

1. Let $f(x) = x^3 + 2x$. Determine:

(a) $f(2)$.

Solution.

$$f(2) = 2^3 + 2(2) = 8 + 4 = \boxed{12}.$$

(b) $f(2x)$.

Solution.

$$f(2x) = (2x)^3 + 2(2x) = \boxed{8x^3 + 4x}.$$

(c) $f(-x)$.

Solution.

$$f(-x) = (-x)^3 + 2(-x) = \boxed{-x^3 - 2x} = -f(x).$$

(d) $f(2 + \Delta x)$.

Solution.

$$\begin{aligned} f(2 + \Delta x) &= (2 + \Delta x)^3 + 2(2 + \Delta x) \\ &= 8 + 12\Delta x + 6(\Delta x)^2 + (\Delta x)^3 + 4 + 2\Delta x \\ &= \boxed{12 + 14\Delta x + 6(\Delta x)^2 + (\Delta x)^3}. \end{aligned}$$

2. Let $u(t) = \frac{1}{2}t^2 + t$. Determine:

(a) $u(-1)$.

Solution.

$$u(-1) = \frac{1}{2}(-1)^2 + (-1) = \frac{1}{2} - 1 = \boxed{-\frac{1}{2}}.$$

(b) $u(t + \Delta t)$.

Solution.

$$\begin{aligned} u(t + \Delta t) &= \frac{1}{2}(t + \Delta t)^2 + (t + \Delta t) \\ &= \frac{1}{2}t^2 + t\Delta t + \frac{1}{2}(\Delta t)^2 + t + \Delta t \\ &= \boxed{\frac{1}{2}t^2 + t + (t + 1)\Delta t + \frac{1}{2}(\Delta t)^2}. \end{aligned}$$

(c) $u(-t)$.

Solution.

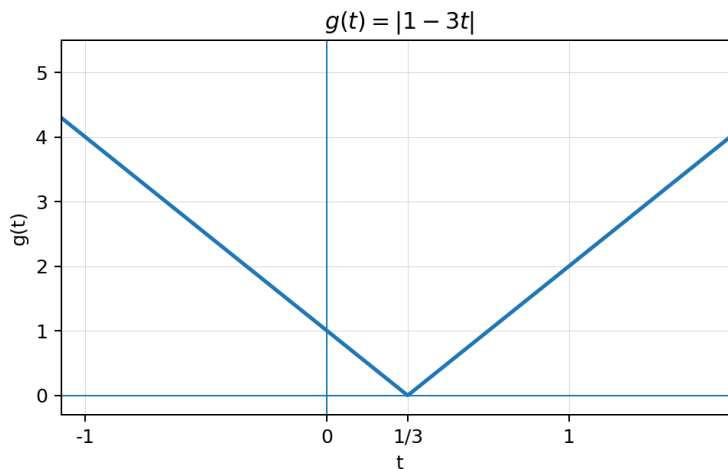
$$u(-t) = \frac{1}{2}(-t)^2 - t = \boxed{\frac{1}{2}t^2 - t}.$$

3. Sketch $g(t) = |1 - 3t|$.

Solution. Since $1 - 3t = 0$ when $t = \frac{1}{3}$,

$$g(t) = \begin{cases} 1 - 3t, & t \leq \frac{1}{3}, \\ 3t - 1, & t \geq \frac{1}{3}. \end{cases}$$

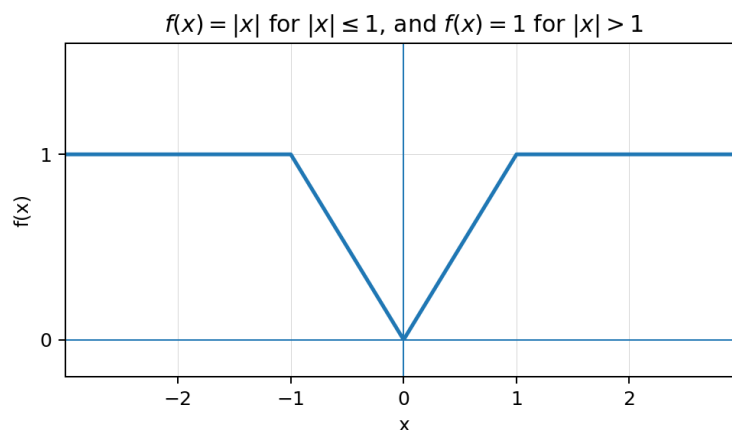
Thus the graph is a V with vertex $(\frac{1}{3}, 0)$, left slope -3 , and right slope 3 .



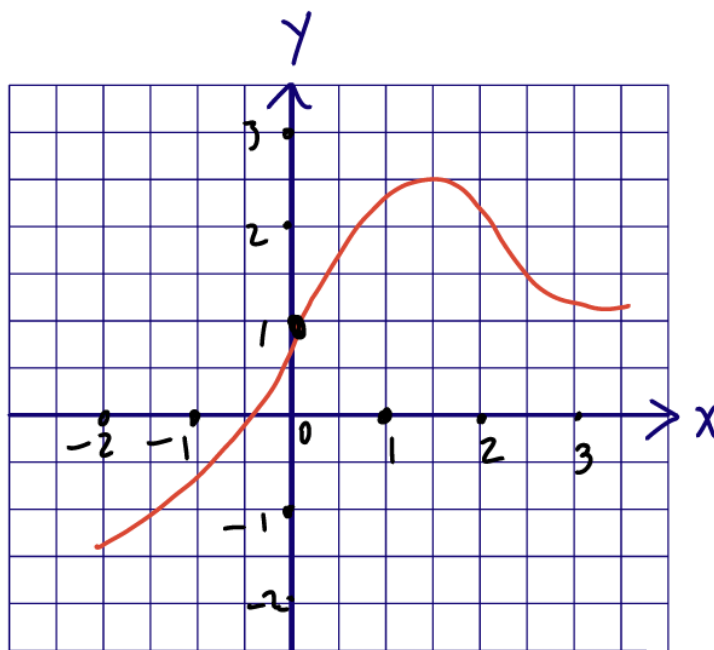
4. Sketch

$$f(x) = \begin{cases} |x|, & |x| \leq 1, \\ 1, & |x| > 1. \end{cases}$$

Solution. On $[-1, 1]$ the graph is the usual V-shaped graph $y = |x|$. Outside that interval the graph is the horizontal line $y = 1$. The pieces meet at $(-1, 1)$ and $(1, 1)$.



5. The graph of f is shown below.



(a) State $f(1.5)$.

Solution. From the graph, $f(1.5) \approx 2.5$.

(b) Estimate $f(-1)$.

Solution. From the graph, $f(-1) \approx -0.6$ (an estimate near -0.5 to -0.6 is reasonable from the hand-drawn graph).

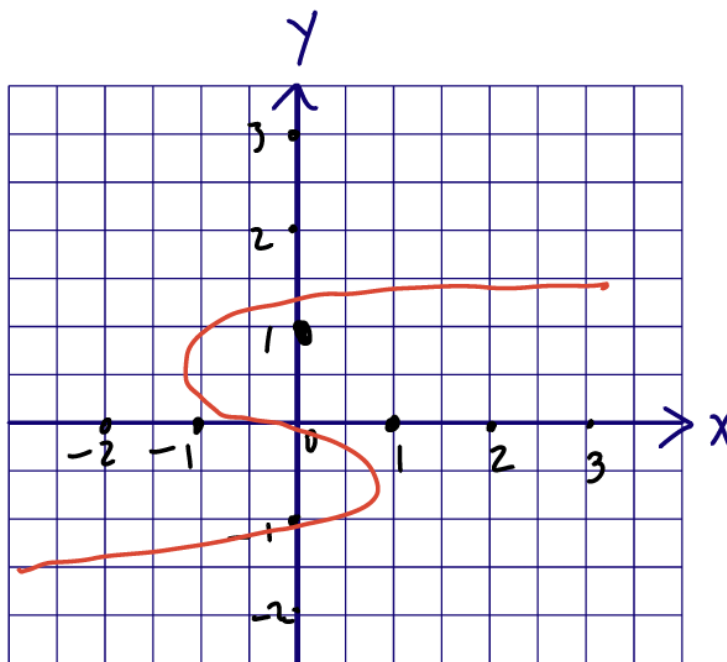
(c) For what values of x is $f(x) = 1$?

Solution. The horizontal line $y = 1$ meets the graph at $x = 0$, so $x = 0$.

(d) Estimate the value of x such that $f(x) = 0$.

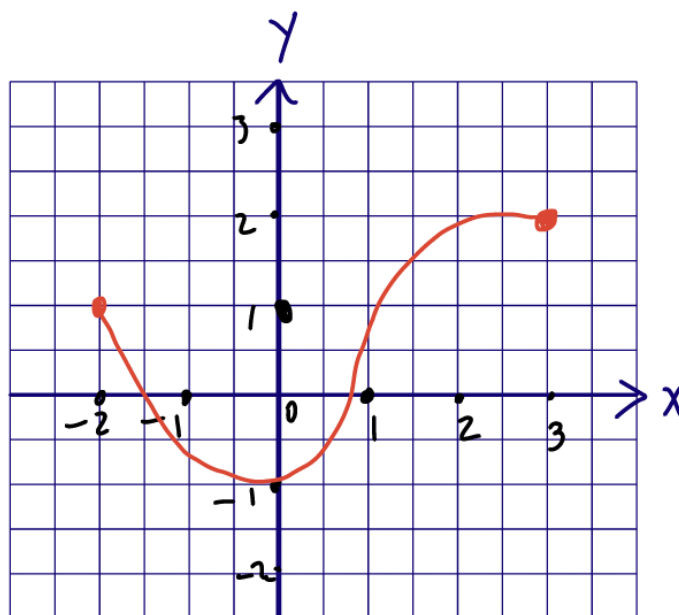
Solution. The x -intercept is approximately $x \approx -0.4$.

6. Determine whether the curve below is the graph of a function of x . If it is, state its domain and range.



Solution. It is not the graph of a function of x . A vertical line through some x -values intersects the red curve more than once, so it fails the vertical line test.

7. State the domain and range of the function graphed below.



Solution. The graph begins at the closed point with $x = -2$ and ends at the closed point with $x = 3$, so

$$\text{Domain} = [-2, 3].$$

Its lowest value is $y = -1$ and its highest value is $y = 2$, so

$$\boxed{\text{Range} = [-1, 2]}.$$

8. Find the domain of

$$f(x) = \frac{x+4}{x^2-9}.$$

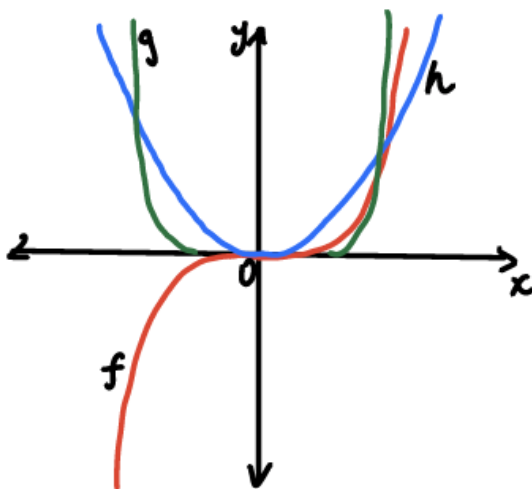
Solution. The denominator cannot equal zero:

$$x^2 - 9 = (x-3)(x+3) \neq 0.$$

Thus $x \neq -3, 3$, and

$$\boxed{\text{Domain} = (-\infty, -3) \cup (-3, 3) \cup (3, \infty)}.$$

9. Match each equation with its graph: (a) $y = x^2$, (b) $y = x^5$, (c) $y = x^8$.



Solution. The red curve f takes negative values for negative x and positive values for positive x , so it is the odd power:

$$\boxed{f(x) = x^5}.$$

The blue curve h is the wider even-power graph, while the green curve g is flatter near 0 and grows more steeply for $|x| > 1$. Therefore

$$\boxed{h(x) = x^2, \quad g(x) = x^8}.$$

So the matching is

$$\boxed{\text{(a) } h, \quad \text{(b) } f, \quad \text{(c) } g}.$$

MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 4

Due **Monday, September 14** at 11:55pm, on Gradescope

GOALS:

- Practice graphing of functions.
- Practice shifting and stretching graphs.
- Simplify difference quotients.
- Practice compositions of functions.

- Write complete solutions.

Justify your solution, writing down the steps that led you to the final answer.

- (1) The graph of $y = f(x)$ is given. Match each equation with its graph and give reasons for your choices.

(a)

$$y = f(x - 4)$$

(b)

$$y = f(x) + 3$$

(c)

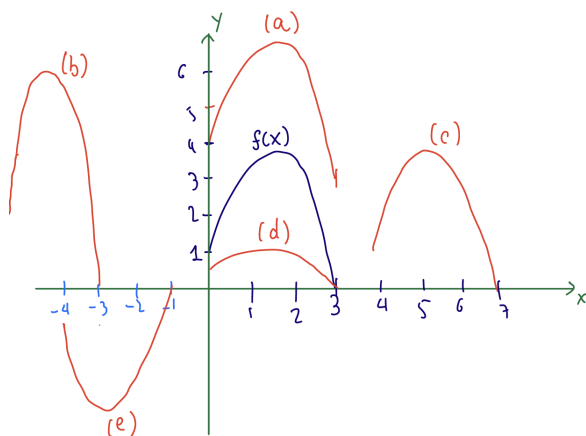
$$y = \frac{1}{3}f(x)$$

(d)

$$y = -f(x + 4)$$

(e)

$$y = 2f(x + 4)$$



- (2) Graph the following functions by hand, not by plotting points, but by starting with the graph of one of the standard functions (see Stewart, Section 1.2) and then applying the appropriate transformations.

(a) $y = -x^2$

(b) $1 - \frac{1}{x}$

(c) $|x - 2|$

- (3) Let $f(x) = \frac{x+1}{x}$.

(a) Compute $f(a + h)$

(b) Simplify

$$\frac{f(a + h) - f(a)}{h}$$

- (4) Find $f(g(h(x)))$:

$$f(x) = \sqrt{x - 3}, g(x) = x^2, h(x) = x^3 + 2$$

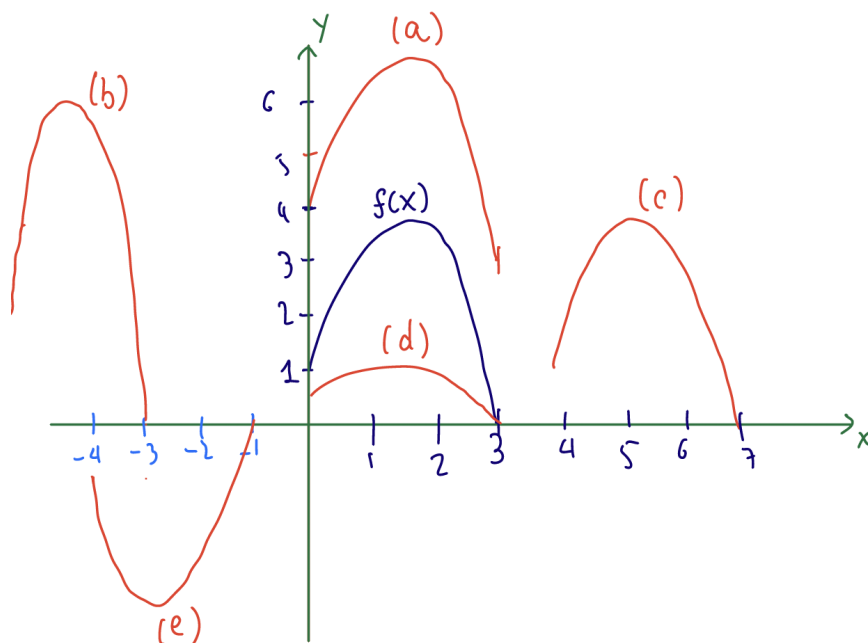
- (5) Express the function in the form $f(g(x))$:

$$F(x) = (2x + x^2)^4$$

(In other words, find functions $f(x)$ and $g(x)$ such that $F(x) = f(g(x))$).

MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 4 SOLUTIONS

- (1) The graph of $y = f(x)$ is given. Match each equation with its graph and give reasons for your choices.

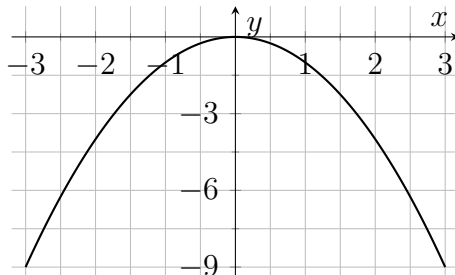


- (a) $y = f(x - 4)$ is f shifted four units to the right, so it is (c).
- (b) $y = f(x) + 3$ is f shifted three units up, so it is (a).
- (c) $y = \frac{1}{3}f(x)$ is a vertical compression by a factor of $\frac{1}{3}$, so it is (d).
- (d) $y = -f(x + 4)$ shifts f four units left and reflects it across the x -axis, so it is (e).
- (e) $y = 2f(x + 4)$ shifts f four units left and stretches it vertically by a factor of 2, so it is (b).

- (2) Graph the following functions by hand, starting with a standard graph and applying the appropriate transformations.

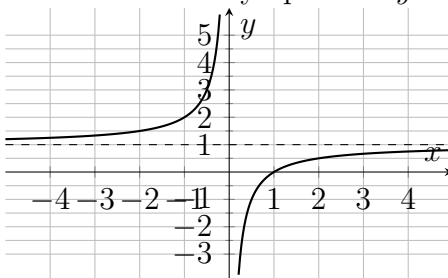
(a) $y = -x^2$

Start with $y = x^2$ and reflect it across the x -axis. The vertex remains at $(0, 0)$ and the parabola opens downward.



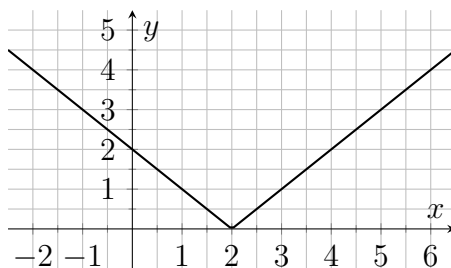
(b) $y = 1 - \frac{1}{x}$

Start with $y = \frac{1}{x}$, reflect across the x -axis, and then shift up by 1. The vertical asymptote is $x = 0$ and the horizontal asymptote is $y = 1$.



(c) $y = |x - 2|$

Start with $y = |x|$ and shift it two units to the right. The vertex is $\boxed{(2, 0)}$.



(3) Let

$$f(x) = \frac{x+1}{x}.$$

(a) Replacing x by $a + h$ gives

$$\boxed{f(a+h) = \frac{a+h+1}{a+h}}.$$

(b) We simplify the difference quotient:

$$\begin{aligned} \frac{f(a+h) - f(a)}{h} &= \frac{1}{h} \left(\frac{a+h+1}{a+h} - \frac{a+1}{a} \right) \\ &= \frac{1}{h} \left(\frac{a(a+h+1) - (a+1)(a+h)}{a(a+h)} \right) \\ &= \frac{1}{h} \left(\frac{-h}{a(a+h)} \right) \\ &= \boxed{-\frac{1}{a(a+h)}}. \end{aligned}$$

The original expression requires $a \neq 0$, $a+h \neq 0$, and $h \neq 0$.

(4) Find $f(g(h(x)))$, where

$$f(x) = \sqrt{x-3}, \quad g(x) = x^2, \quad h(x) = x^3 + 2.$$

Work from the inside out:

$$h(x) = x^3 + 2,$$

so

$$g(h(x)) = (x^3 + 2)^2.$$

Therefore

$$f(g(h(x))) = \sqrt{(x^3 + 2)^2 - 3}.$$

Thus

$$\boxed{f(g(h(x))) = \sqrt{(x^3 + 2)^2 - 3}}.$$

(5) Express

$$F(x) = (2x + x^2)^4$$

in the form $f(g(x))$.

One natural choice is

$$\boxed{g(x) = 2x + x^2} \quad \text{and} \quad \boxed{f(x) = x^4}.$$

Indeed,

$$f(g(x)) = (g(x))^4 = (2x + x^2)^4 = F(x).$$

Other choices are possible.

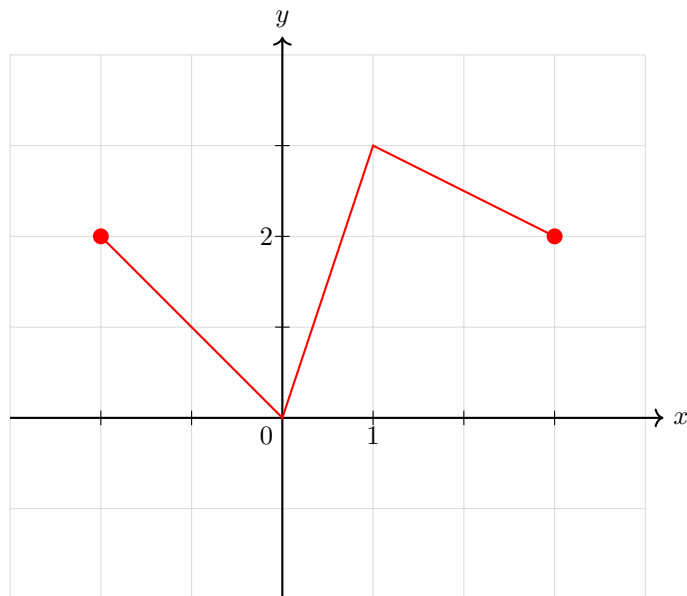
MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 5

Due **Friday, September 18** at 11:55pm, on Gradescope

Justify your solutions, writing down the steps that led you to the final answer.

(1) The graph of f is given. Draw the graphs of the following functions.

- (a) $y = f(x) - 3$
- (b) $y = f(x + 1)$
- (c) $y = \frac{1}{2}f(x)$
- (d) $y = -f(x)$
- (e) $y = f + f$.



(2) Graph the function $f(x) = -1 + \frac{1}{(2x)^2 - 1}$

(3) Let $f(x) = 1 - x^2$ and let $g(x) = \sqrt{x + 1}$. Compute (and simplify) $f + g$, fg , $f \circ g$, and $g \circ f$.

(4) Explain in your own words what is meant by the equation

$$\lim_{x \rightarrow 2} f(x) = 5.$$

Is it possible for this statement to be true and yet $f(2) = 3$? Explain.

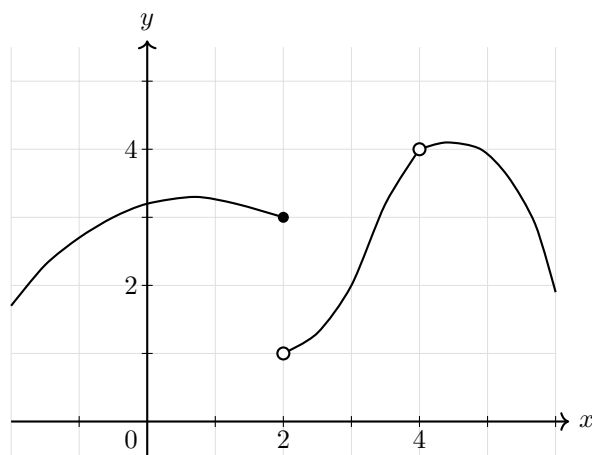
- (5) Explain what it means to say that

$$\lim_{x \rightarrow 1^-} f(x) = 3 \quad \text{and} \quad \lim_{x \rightarrow 1^+} f(x) = 7.$$

In this situation is it possible that $\lim_{x \rightarrow 1} f(x)$ exists? Explain.

- (6) For the function f whose graph is given below, state the value of each quantity, if it exists. If it does not exist, explain why.

- (a) $\lim_{x \rightarrow 2^-} f(x)$
- (b) $\lim_{x \rightarrow 2^+} f(x)$
- (c) $\lim_{x \rightarrow 2} f(x)$
- (d) $f(2)$
- (e) $\lim_{x \rightarrow 4} f(x)$
- (f) $f(4)$
- (g) $\lim_{x \rightarrow 3} f(x)$



- (7) Determine the limits:

$$\lim_{x \rightarrow 5^+} \frac{x+1}{x-5}.$$

and

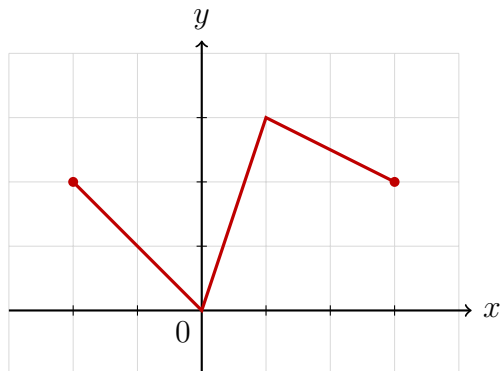
$$\lim_{x \rightarrow 5^-} \frac{x+1}{x-5}.$$

MATH 105 CALCULUS WITH ALGEBRA – FALL 2026
HOMEWORK 5 SOLUTIONS

(1) The graph of f is given by the piecewise-linear curve through

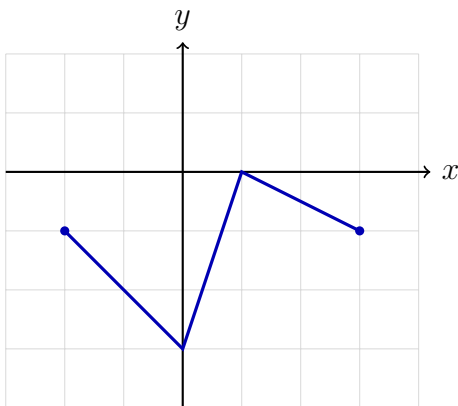
$$(-2, 2), \quad (0, 0), \quad (1, 3), \quad (3, 2).$$

We transform the entire graph, rather than transforming the individual points one at a time.



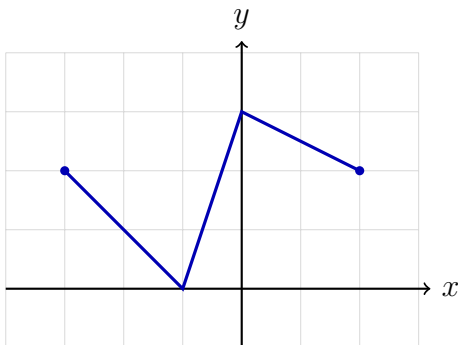
(a) $y = f(x) - 3$ shifts the graph of f **down 3 units**. The transformed vertices are

$$(-2, -1), \quad (0, -3), \quad (1, 0), \quad (3, -1).$$



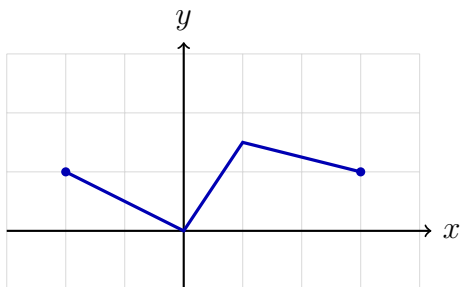
(b) $y = f(x + 1)$ shifts the graph of f **left 1 unit**. The transformed vertices are

$$(-3, 2), \quad (-1, 0), \quad (0, 3), \quad (2, 2).$$



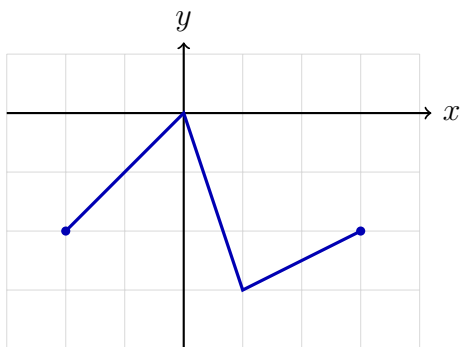
- (c) $y = \frac{1}{2}f(x)$ vertically compresses the graph by a factor of $\frac{1}{2}$. The x -coordinates stay the same, while all y -coordinates are halved:

$$(-2, 1), \quad (0, 0), \quad \left(1, \frac{3}{2}\right), \quad (3, 1).$$



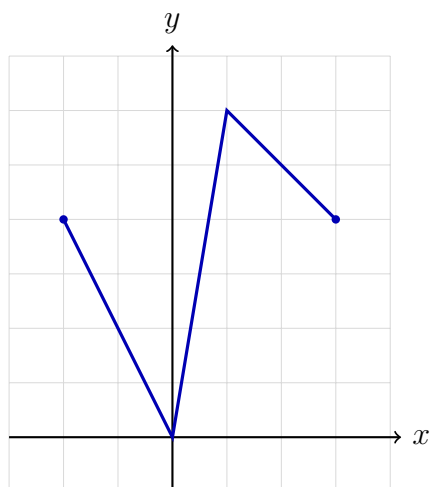
- (d) $y = -f(x)$ reflects the graph across the x -axis. The transformed vertices are

$$(-2, -2), \quad (0, 0), \quad (1, -3), \quad (3, -2).$$



- (e) Since $f + f = 2f$, the graph is a vertical stretch by a factor of 2. The transformed vertices are

$$(-2, 4), \quad (0, 0), \quad (1, 6), \quad (3, 4).$$



(2) Graph

$$f(x) = -1 + \frac{1}{(2x)^2 - 1} = -1 + \frac{1}{4x^2 - 1}.$$

The denominator is zero when

$$4x^2 - 1 = 0,$$

so the vertical asymptotes are

$$\boxed{x = -\frac{1}{2} \quad \text{and} \quad x = \frac{1}{2}}.$$

As $|x|$ becomes large, the fraction $1/(4x^2 - 1)$ approaches 0, so the horizontal asymptote is

$$\boxed{y = -1}.$$

The function is even, so its graph is symmetric about the y -axis. Also,

$$f(0) = -2.$$

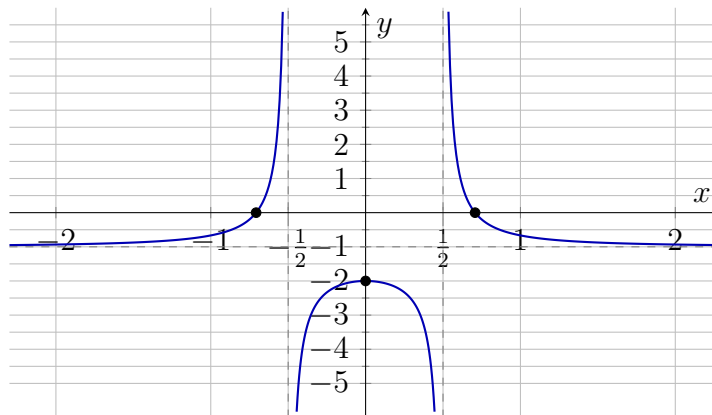
For the x -intercepts,

$$\begin{aligned} -1 + \frac{1}{4x^2 - 1} &= 0 \\ \frac{1}{4x^2 - 1} &= 1 \\ 4x^2 &= 2, \end{aligned}$$

so

$$\boxed{x = \pm \frac{1}{\sqrt{2}}}.$$

These features determine the graph:



(3) Let

$$f(x) = 1 - x^2, \quad g(x) = \sqrt{x + 1}.$$

$$\begin{aligned} (f + g)(x) &= f(x) + g(x) \\ &= \boxed{1 - x^2 + \sqrt{x + 1}}. \end{aligned}$$

This is defined for $x \geq -1$.

For the product,

$$(fg)(x) = \boxed{(1 - x^2)\sqrt{x + 1}},$$

again for $x \geq -1$.

For the first composition,

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\ &= 1 - \left(\sqrt{x+1}\right)^2 \\ &= \boxed{-x}.\end{aligned}$$

Although the simplified formula is $-x$, the original composition requires $x \geq -1$ because $g(x)$ must be defined.

For the other composition,

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) \\ &= \sqrt{(1-x^2)+1} \\ &= \boxed{\sqrt{2-x^2}}.\end{aligned}$$

For this to be defined we need

$$2 - x^2 \geq 0,$$

so its domain is $[-\sqrt{2}, \sqrt{2}]$.

(4) The statement

$$\lim_{x \rightarrow 2} f(x) = 5$$

means that the values of $f(x)$ can be made as close to 5 as we want by taking x sufficiently close to 2 (but with $x \neq 2$).

Yes, it is possible that $f(2) = 3$. A limit describes what happens to $f(x)$ *near* $x = 2$; it does not require the value at $x = 2$ itself to equal the limit.

(5) The statement

$$\lim_{x \rightarrow 1^-} f(x) = 3$$

means that $f(x)$ approaches 3 as x approaches 1 using values with $x < 1$. Similarly,

$$\lim_{x \rightarrow 1^+} f(x) = 7$$

means that $f(x)$ approaches 7 as x approaches 1 using values with $x > 1$.

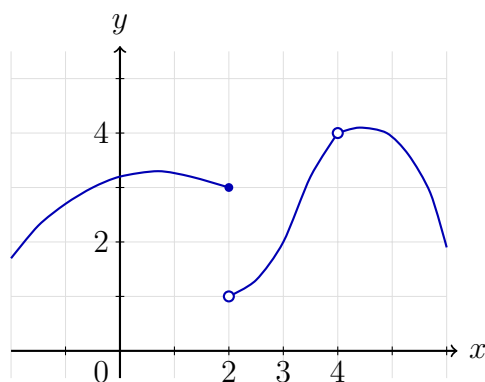
The two-sided limit **does not exist**, because a two-sided limit exists only when the left-hand and right-hand limits agree. Here

$$3 \neq 7.$$

Thus

$$\boxed{\lim_{x \rightarrow 1} f(x) \text{ does not exist.}}$$

(6) For the graph below:



(a) As x approaches 2 from the left, the graph approaches $y = 3$, so

$$\lim_{x \rightarrow 2^-} f(x) = 3.$$

(b) As x approaches 2 from the right, the graph approaches the open point at $(2, 1)$, so

$$\lim_{x \rightarrow 2^+} f(x) = 1.$$

(c) Since the one-sided limits are different,

$$\lim_{x \rightarrow 2} f(x) \text{ does not exist.}$$

(d) The filled point at $x = 2$ is $(2, 3)$, so

$$f(2) = 3.$$

(e) From both sides of $x = 4$, the graph approaches the hole at $(4, 4)$, so

$$\lim_{x \rightarrow 4} f(x) = 4.$$

(f) There is no filled point at $x = 4$, so

$$f(4) \text{ is undefined.}$$

(g) The graph is continuous near $x = 3$ and passes through $(3, 2)$, so

$$\lim_{x \rightarrow 3} f(x) = 2.$$

(7) Consider

$$\frac{x+1}{x-5}.$$

As $x \rightarrow 5$, the numerator approaches 6, which is positive. The sign of the denominator determines the sign of the quotient.

For $x \rightarrow 5^+$, we have $x - 5 \rightarrow 0^+$, so

$$\lim_{x \rightarrow 5^+} \frac{x+1}{x-5} = +\infty.$$

For $x \rightarrow 5^-$, we have $x - 5 \rightarrow 0^-$, so

$$\lim_{x \rightarrow 5^-} \frac{x+1}{x-5} = -\infty.$$

MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 6

Due **Monday, September 21** at 11:55pm, on Gradescope

- (1) Factor the following expressions:

(a)

$$xy + 4x + 2xw$$

(b)

$$10x^8y^6 + 25x^2y^4 + 20x^3y^{10}$$

- (2) Explain in your own words what is meant by the equation

$$\lim_{x \rightarrow 2} f(x) = 5$$

Is it possible for this statement to be true and yet $f(2) = 3$?

- (3) Explain what it means to say that

$$\lim_{x \rightarrow 1^-} f(x) = 3 \text{ and } \lim_{x \rightarrow 1^+} f(x) = 7$$

In this situation is it possible that $\lim_{x \rightarrow 1} f(x)$ exists?

- (4) For the function f whose graph is given, state the value of each quantity, if it exists. If it does not exist, explain why.

(a) $\lim_{x \rightarrow 2^-} f(x)$

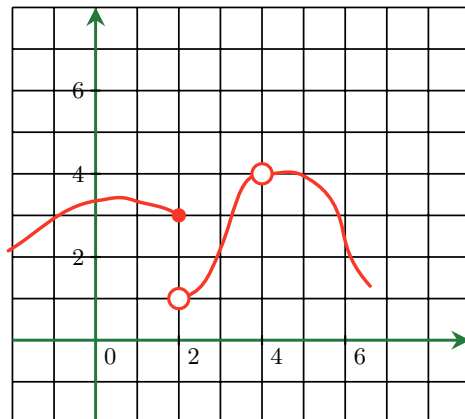
(c) $\lim_{x \rightarrow 2} f(x)$

(e) $\lim_{x \rightarrow 4} f(x)$

(b) $\lim_{x \rightarrow 2^+} f(x)$

(d) $f(2)$

(f) $f(4)$



- (5) For the function g whose graph is given, state the value of each quantity, if it exists. If it does not exist, explain why.

(a) $\lim_{t \rightarrow 0^-} g(t)$

(b) $\lim_{t \rightarrow 0^+} g(t)$

(c) $\lim_{t \rightarrow 0} g(t)$

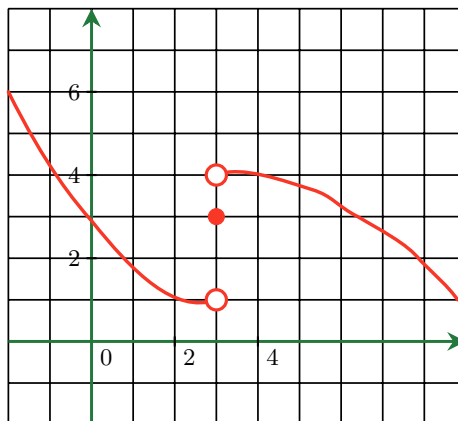
(d) $\lim_{t \rightarrow 3^-} g(t)$

(e) $\lim_{t \rightarrow 3^+} g(t)$

(f) $\lim_{t \rightarrow 3} g(t)$

(g) $g(3)$

(h) $\lim_{t \rightarrow 2} g(t)$



- (6) Sketch the graph of the function and use it to determine the values of a for which $\lim_{x \rightarrow a} f(x)$ exists.

$$f(x) = \begin{cases} 1 + x & \text{if } x < -1 \\ x^2 & \text{if } -1 \leq x < 1 \\ 2 - x & \text{if } x \geq 1 \end{cases}$$

- (7) Guess the value of the limit (if it exists) by evaluating at the given numbers (using a calculator), correct to six decimal places.

$$\lim_{x \rightarrow 3} \frac{x^2 - 3x}{x^2 - 9},$$

where

$$x = 3.1, 3.05, 3.001, 2.9, 2.95, 2.995.$$

Note: I will not ask you to do this on quizzes/exams since calculators won't be available

MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 6 SOLUTIONS

- (1) Factor the following expressions.
(a) Every term has a factor of x , so

$$xy + 4x + 2xw = x(y + 4 + 2w).$$

- (b) The greatest common factor is $5x^2y^4$, so

$$10x^8y^6 + 25x^2y^4 + 20x^3y^{10} = 5x^2y^4(2x^6y^2 + 5 + 4xy^6).$$

- (2) Explain in your own words what is meant by the equation

$$\lim_{x \rightarrow 2} f(x) = 5.$$

Solution. This means that when x is close to 2 (but not necessarily equal to 2), the values of $f(x)$ are close to 5. The closer x gets to 2, the closer $f(x)$ gets to 5.

Yes, it is possible to have $\lim_{x \rightarrow 2} f(x) = 5$ while $f(2) = 3$. A limit describes what happens to $f(x)$ *near* $x = 2$; it does not require the value of the function at $x = 2$ itself to equal the limit.

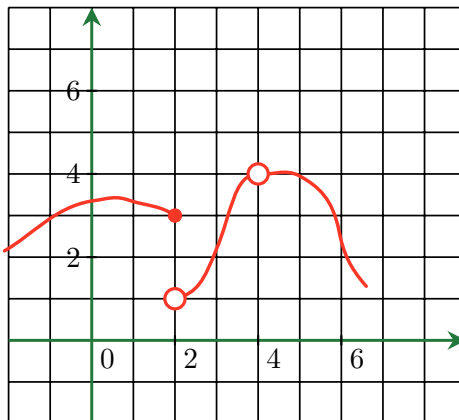
- (3) Explain what it means to say that

$$\lim_{x \rightarrow 1^-} f(x) = 3 \quad \text{and} \quad \lim_{x \rightarrow 1^+} f(x) = 7.$$

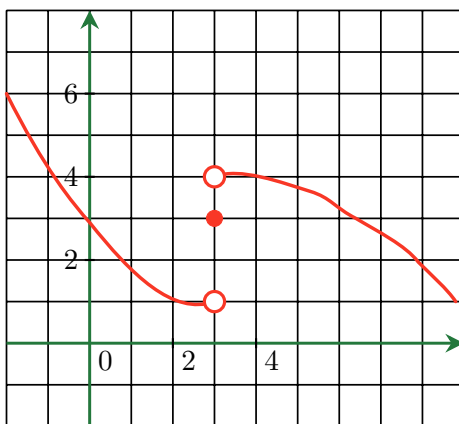
Solution. The first statement means that as x approaches 1 through values less than 1, $f(x)$ approaches 3. The second means that as x approaches 1 through values greater than 1, $f(x)$ approaches 7.

Since the left-hand and right-hand limits are different, the two-sided limit $\lim_{x \rightarrow 1} f(x)$ does not exist.

- (4) For the function f whose graph is given, state the value of each quantity, if it exists.



- (a) As $x \rightarrow 2^-$, the graph approaches the filled point at height 3, so $\lim_{x \rightarrow 2^-} f(x) = 3$.
- (b) As $x \rightarrow 2^+$, the graph approaches the open point at height 1, so $\lim_{x \rightarrow 2^+} f(x) = 1$.
- (c) Since the two one-sided limits are different, $\lim_{x \rightarrow 2} f(x)$ does not exist.
- (d) The filled point at $x = 2$ is at height 3, so $f(2) = 3$.
- (e) From both sides of $x = 4$, the graph approaches the open point at height 4, so $\lim_{x \rightarrow 4} f(x) = 4$.
- (f) There is no filled point on the graph at $x = 4$, so $f(4)$ is undefined.
- (5) For the function g whose graph is given, state the value of each quantity, if it exists.



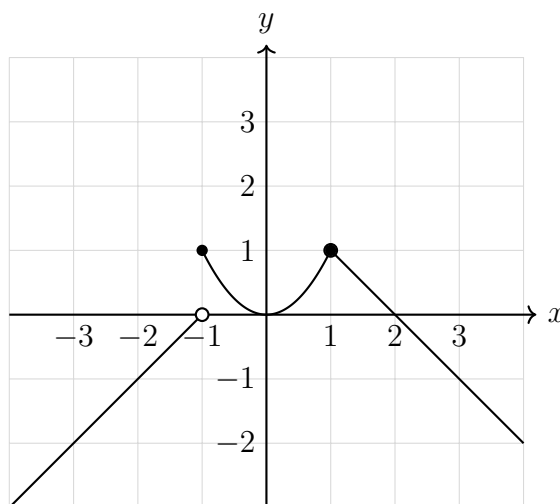
- (a) The graph is continuous near $t = 0$ and is at about height 3, so $\lim_{t \rightarrow 0^-} g(t) \approx 3$.
- (b) Similarly, $\lim_{t \rightarrow 0^+} g(t) \approx 3$.
- (c) Since the one-sided limits agree, $\lim_{t \rightarrow 0} g(t) \approx 3$.
- (d) Approaching $t = 3$ from the left, the graph approaches the open point at height 1, so $\lim_{t \rightarrow 3^-} g(t) = 1$.

- (e) Approaching $t = 3$ from the right, the graph approaches the open point at height 4, so $\lim_{t \rightarrow 3^+} g(t) = 4$.
- (f) Since the two one-sided limits are different, $\lim_{t \rightarrow 3} g(t)$ does not exist.
- (g) The filled point at $t = 3$ is at height 3, so $g(3) = 3$.
- (h) The graph is continuous near $t = 2$ and is at about height 1, so $\lim_{t \rightarrow 2} g(t) \approx 1$.

(6) Sketch the graph of

$$f(x) = \begin{cases} 1 + x & \text{if } x < -1, \\ x^2 & \text{if } -1 \leq x < 1, \\ 2 - x & \text{if } x \geq 1. \end{cases}$$

Solution. For $x < -1$, the graph is the line $y = x + 1$, with an open point at $(-1, 0)$. For $-1 \leq x < 1$, it is the parabola $y = x^2$, with a closed point at $(-1, 1)$ and an open point at $(1, 1)$. For $x \geq 1$, it is the line $y = 2 - x$, beginning with a closed point at $(1, 1)$.



At $x = -1$, the left-hand limit is 0 while the right-hand limit is 1, so the limit does not exist there. At $x = 1$, both sides approach 1, so the limit does exist. Everywhere else, the function is given by a polynomial or a line and has no break.

Therefore $\lim_{x \rightarrow a} f(x)$ exists for every real number a except $a = -1$; that is, for

$$a \in \mathbb{R} \setminus \{-1\}.$$

(7) Guess the value of

$$\lim_{x \rightarrow 3} \frac{x^2 - 3x}{x^2 - 9}$$

by evaluating the expression at the given values of x .

Solution. Using a calculator and rounding to six decimal places gives

x	$\frac{x^2 - 3x}{x^2 - 9}$
3.1	0.508197
3.05	0.504132
3.001	0.500083
2.9	0.491525
2.95	0.495798
2.995	0.499583

The values from both sides of 3 are getting closer to 0.5. Thus the data suggest that

$$\lim_{x \rightarrow 3} \frac{x^2 - 3x}{x^2 - 9} = \frac{1}{2}.$$

MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 7

Due **Friday, September 25** at 11:55pm, on Gradescope

(1) Determine the following limits.

(a) $\lim_{x \rightarrow 1} \frac{2-x}{(x-1)^2}$

(c) $\lim_{x \rightarrow 5^+} \frac{x^2 - 25}{|x - 5|}$

(b) $\lim_{x \rightarrow -2^+} \frac{x-1}{x^2(x+2)}$

(d) $\lim_{x \rightarrow 1} \frac{1 - \sqrt{x}}{x - 5}$

(2) Suppose that f and g are functions such that

$$\lim_{x \rightarrow 2} f(x) = 4 \quad \lim_{x \rightarrow 2} g(x) = -2 \quad \lim_{x \rightarrow 2} h(x) = 0$$

Which of the below limits exist, and if they exist, what are their values? Explain your work.

(a) $\lim_{x \rightarrow 2} (f(x) + 5g(x))$

(c) $\lim_{x \rightarrow 2} \sqrt{f(x)}$

(b) $\lim_{x \rightarrow 2} (g(x))^3$

(d) $\lim_{x \rightarrow 2} \frac{g(x)}{h(x)}$

(3) The *signum* (or sign) *function*, denoted by sgn , is defined by

$$\text{sgn} = \begin{cases} -1 & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ 1 & \text{if } x > 0 \end{cases}$$

(a) Sketch the graph of this function.

(b) Find each of the following limits or explain why it does not exist.

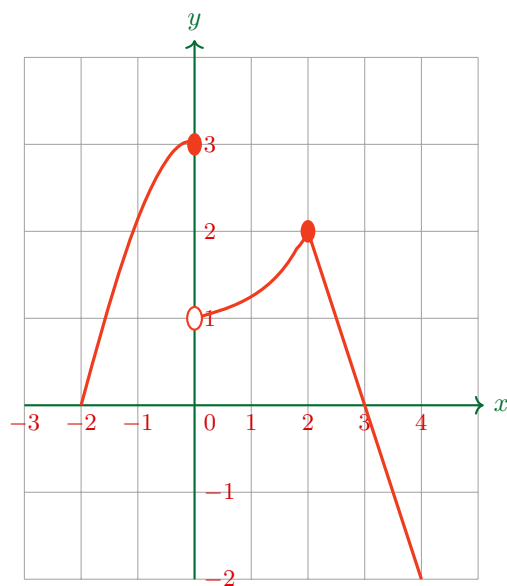
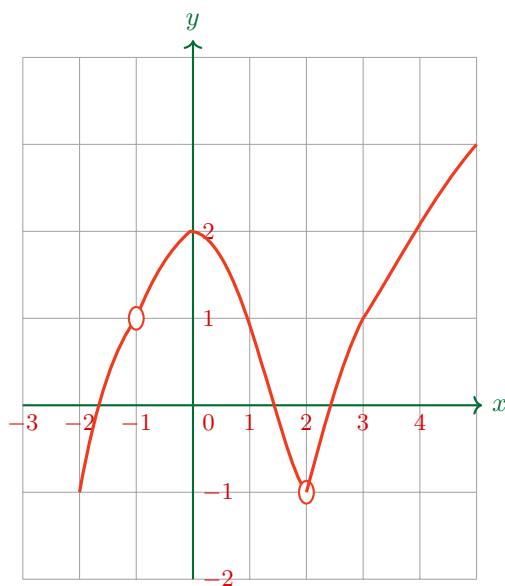
(i) $\lim_{x \rightarrow 0^+} \text{sgn } x$

(iii) $\lim_{x \rightarrow 0} \text{sgn } x$

(ii) $\lim_{x \rightarrow 0^-} \text{sgn } x$

(iv) $\lim_{x \rightarrow 0} |\text{sgn } x|$

- (4) The graphs of f and g are given. Use them to evaluate each limit, if it exists. If the limit does not exist, explain why.



(i) $\lim_{x \rightarrow 2} [f(x) + g(x)]$

(ii) $\lim_{x \rightarrow 2} [f(x) - g(x)]$

(iii) $\lim_{x \rightarrow -1} [f(x)g(x)]$

(iv) $\lim_{x \rightarrow 2} [x^2 f(x)]$